

Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

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Research Experience

University of Wyoming, Graduate Research Assistant

Laramie, WY, USA
2025 – present

- Built an axisymmetric finite-volume model (OpenFOAM + solids4Foam) of a spherical proppant grain loaded into and unloaded from Mohr–Coulomb rock through penalty contact; unloading separates permanent embedment from elastic recovery. All rock inputs are published Montney siltstone measurements; nothing is fitted to indentation data
- Compared the model with published Brinell indentation (35 N) and with eight proppant-pack embedment tests predicted blind; the pack comparison shows a grain-size-dependent mismatch that neither mesh refinement nor dilation angle removes, reported as a model limit rather than tuned away
- Showed that residual depth converges slowly with mesh and that the earlier training set carried a mesh bias; now re-running every condition on three mesh levels with Richardson extrapolation and uncertainty bands (≈ 750 runs on UW's MedicineBow cluster, Oct. 2026)
- Fitting a dimensionless residual-depth equation (load, cohesion, friction and dilation angles, grain/rock stiffness) by nested leave-family-out cross-validation; added 60 conditions chosen by a pre-registered balance rule so the equation's range of validity can be stated factor by factor
- Pre-registered the test matrix and pass/fail checks (convergence, force balance) in Git before running, and automated the post-processing so every figure and statistic regenerates from the archived runs
- Trained in triaxial cell operation and DCI rock-compressibility (bulk modulus) testing
- Planned next: inverse models with prediction intervals to find which indentation measurement identifies the dilation angle; preparing the indentation experiments that will supply the data
- Designed and built a displacement-controlled micro-indentation rig for P–h curves on rock and metal (parametric CAD, 3D-printed fixtures, 1 mm tungsten-carbide ball) with a piezoelectric load cell and a laser Doppler vibrometer recorded through an NI DAQ in LabVIEW

KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time)

South Korea
2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Sole investigator on a government-funded study of air-jet-driven waves on a thin elastic film: derived the dispersion relation with film stiffness and tension and confirmed the Kelvin-wave transition on high-speed video
- Assembled a free-surface cavitation tank from parts and explained the upper-wall contraction of ventilated supercavities with a wake-theory term in a 2D model's closure condition, closing most of the gap to the photos
- Simulated von Kármán vortex shedding near a free surface (ANSYS Fluent, VOF): the closer the surface, the lower the Strouhal number and the weaker the upper vortices

Pusan National University — Department of Naval Architecture & Ocean Engineering, Undergraduate Research Assistant - Validated RANS CFD of KCS hull resistance against the Gothenburg 2010 benchmark (STAR-CCM+): 1.2% error in total resistance; responsible for mesh refinement at the bow, stern and free surface and for post-processing - PIV and hull-resistance tests, ITTC 1957/1978 propulsion design, and an ANSYS Fluent study of an airfoil appendage on hull attitude and resistance - Started the vibration/acoustic anomaly-detection model (MATLAB, Python) finished at Samsung	South Korea 2022 – 2024
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Industry Experience

Samsung Heavy Industries Laboratory, Mechanical Engineer Intern Trained an LSTM autoencoder on healthy-condition vibration and acoustic data only (Green Nuri vessel program) so anomalies appear as reconstruction error; detected 621 of 622 test anomalies, one traced to a weld fatigue crack	South Korea 2024 – 2024
Pusan National University, Capstone Design Project - Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria - Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability	South Korea 2023 – 2023

Publications

Development of a dimensionless predictive equation for residual proppant embedment from finite-volume simulations of Montney siltstone Seungmin Yome, Soheil Saraji (in preparation)	Jan 2026
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Skills

Simulation & CAE

Programming

Design & Engineering Tools

Laboratory

Education

University of Wyoming, Energy & Petroleum Engineering - Advisor: Dr. Soheil Saraji (Department Head) - Minor: Computer Science (in progress) - Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam - Coursework: Rock Mechanics, Enhanced Oil Recovery, Geostatistics, Python for Engineers - Current coursework (Fall 2026): ML for Computational Solid Mechanics, Practical Machine Learning	Laramie, WY, USA 2025 – present
Pusan National University, Naval Architecture & Ocean Engineering - Advisor: Dr. Inwon Lee - Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD	South Korea 2018 – 2024

Grants & Fellowships

Graduate Research Assistant, University of Wyoming (fully funded) University of Wyoming	2025–Present
Research Scholarship Korea Advanced Institute of Science and Technology	Jan 2024
Undergraduate Research Student Grants Pusan National University	2022–2024
Academic Excellence Scholarship Pusan National University	2022–2024
Foreign Language Special Lecture Consignment Program Scholarship Pusan National University	Jan 2021
Creative Human Resources Development Project Group Scholarship Pusan National University	Jan 2018

Selected Activities & Awards

Invited to Samsung Heavy Industries Wind Propulsion Ship Forum (travel award) Samsung Heavy Industries	July 2024
Participating in ‘Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship’ Korean Society of Shipbuilding and Marine Engineering / Coast Guard	Jan. 2024
Invited to Hyundai Heavy Industries Eco-Friendly Smart Ship Forum (travel award) Hyundai Heavy Industries	Apr. 2023
TPC (Test Preparation Course) Certification — English Consignment Program Pusan National University	July 2021

Teaching & Tutoring

Private Tutoring (Educational Volunteering), Mathematics Instructor Taught high-school mathematics as an educational volunteer for two years	South Korea 2022 – 2024
Private Tutoring, Physics & Chemistry Instructor — College Entrance Preparation Taught physics and chemistry to ten high-school seniors preparing for the Korean college entrance exam (CSAT) over six months	South Korea 2023 – 2024

Languages

Korean
Native

English
Fluent